CLINICAL · NECK, THYROID & SALIVARY

Salivary gland tumours

Salivary tumours follow a simple rule of size: the bigger the gland, the more likely its tumour is benign. Four out of five parotid tumours are benign, about half of submandibular tumours, and only a minority of minor salivary and sublingual tumours. Most parotid lumps are pleomorphic adenomas, slow and painless, that must be removed with a cuff of normal gland because they leave seedlings behind if shelled out. Pain, rapid growth, fixity, nodes and above all facial weakness mean malignancy.

AT A GLANCE

- 80% of parotid tumours are benign; most minor gland tumours are malignant.
- Pleomorphic adenoma: pseudopodia, so never enucleate.
- Warthin: tail of parotid, older male smokers, bilateral in some.
- Adenoid cystic carcinoma spreads along nerves and to the lungs.
- Permanent facial weakness about 7% after parotidectomy; Frey syndrome about 20%; Milan system for cytology.

1 CHAPTER ONE Overview

Classification and clues.

1.1 Proportions

- **Parotid:** about 80% of all salivary tumours; about 80% benign.
- **Submandibular:** about 10%; 50 to 60% benign.
- **Minor salivary and sublingual:** the remainder; most minor salivary tumours (and nearly all sublingual tumours) are malignant.

TUMOURS OF THE SALIVARY GLANDS (SIMPLIFIED WHO)

BENIGN	MALIGNANT
Pleomorphic adenoma	Mucoepidermoid carcinoma (low, intermediate, high grade)

BENIGN	MALIGNANT
Warthin tumour (papillary cystadenoma lymphomatosum)	Adenoid cystic carcinoma
Oncocytoma	Acinic cell carcinoma
Basal cell, canalicular adenoma; myoepithelioma	Carcinoma ex pleomorphic adenoma
Haemangioma, lymphangioma (children)	Salivary duct carcinoma, adenocarcinoma, polymorphous adenocarcinoma
Lipoma, neurofibroma, schwannoma	Epithelial–myoepithelial, secretory carcinoma
	Squamous and undifferentiated carcinoma (often metastatic), lymphoma, sarcoma

1.2 Features of malignancy

Rapid growth, pain, fixation to skin or deep structures, ulceration, *facial nerve weakness*, trismus, and cervical nodes. Many malignant tumours, however, present as a painless mobile lump exactly like an adenoma.

1.3 Evaluation

- **Ultrasound** with **fine-needle aspiration** (or core biopsy): about 90% accurate in separating benign from malignant; incisional biopsy of a parotid mass is avoided because it seeds tumour and endangers the nerve.
- **MRI:** the best study for extent, deep lobe and parapharyngeal extension, and perineural spread (especially of adenoid cystic carcinoma along the facial and trigeminal nerves); **CT** for bone erosion; **PET-CT** for high-grade cancers.
- **Facial nerve** function recorded before surgery.

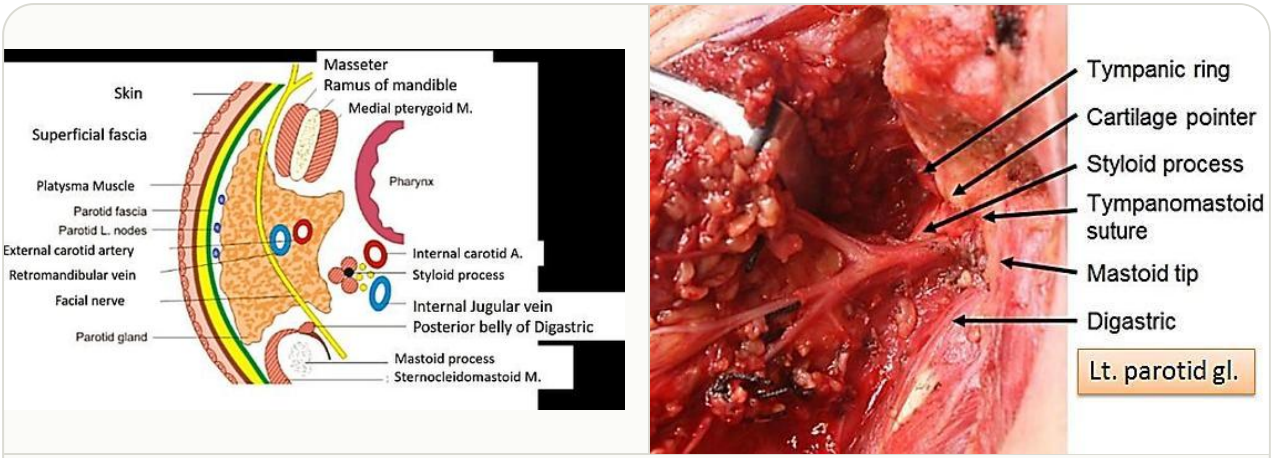


FIGURE 1 The parotid in cross-section with its relations, and the surgical landmarks of the facial nerve at parotidectomy.

Anatomy atlas · ENT Steps p. 182, ENT Steps p. 187

CHAPTER TWO

2 Benign tumours

Adenomas and others.

2.1 Pleomorphic adenoma

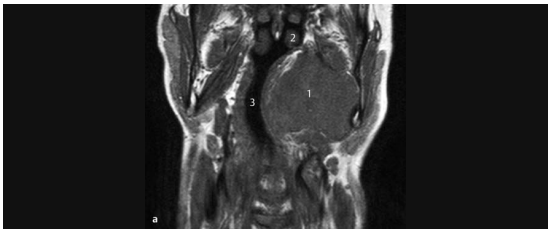
The commonest salivary tumour in adults and children; about 85% in the parotid, usually the superficial lobe or tail; also the submandibular gland and palate. Women more than men, usually in the third to fifth decades. A slow, painless, firm, mobile, lobulated lump that may grow very large; from the deep lobe it can bulge into the parapharyngeal space and push the tonsil and soft palate medially.

- **Pathology:** a *mixed tumour* of epithelial and myoepithelial cells in a myxoid, chondroid or fibrous stroma, within an *incomplete capsule* from which finger-like pseudopodia protrude.
- **Treatment:** *superficial parotidectomy* (or partial superficial parotidectomy) with identification and preservation of the facial nerve, removing the tumour with a cuff of normal tissue; extracapsular dissection is used by some for small, superficial, mobile tumours. *Enucleation* leaves pseudopodia and gives high, often multinodular recurrence; spillage at surgery also seeds recurrence. Deep lobe and parapharyngeal tumours are removed through the neck (occasionally with mandibulotomy), not through the mouth.

- **Malignant change:** *carcinoma ex pleomorphic adenoma* in long-standing or recurrent tumours (risk rising with time, several per cent at ten to fifteen years): sudden growth, pain or facial weakness in an old lump.

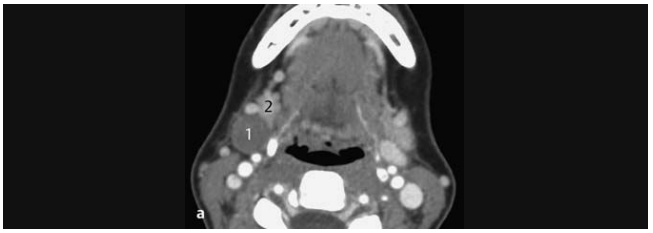
IMAGING IN THE APP 2 cases

Pleomorphic adenoma of the parotid and submandibular glands.



Pleomorphic Adenoma of the Parotid Gland

Tumours · 11 images



Pleomorphic Adenoma of the Submandibular Gland

Tumours · 2 images

2.2 Warthin tumour

Papillary cystadenoma lymphomatosum: the second commonest benign parotid tumour, almost always in the tail of the parotid, in older men who smoke; up to a fifth multifocal and about 5 to 10% bilateral. A soft or cystic, sometimes fluctuating, lump that may become inflamed and painful. Histologically oncocytic epithelium in papillary cysts over a lymphoid stroma; it takes up technetium pertechnetate (a "hot" nodule). Not a true neoplasm and very rarely malignant: removal by partial parotidectomy or extracapsular excision, or observation in the elderly once FNA is confident.

2.3 Oncocytoma

Rare (about 1%), from oncocytes rich in mitochondria, in the elderly, in the superficial parotid, usually under 5 cm; hot on pertechnetate scanning. Superficial parotidectomy.

2.4 Other benign tumours

Basal cell adenoma (can mimic adenoid cystic carcinoma), canalicular adenoma (upper lip), myoepithelioma, lipoma (characteristic on imaging), and neural tumours.

2.5 Tumours in children

Vascular lesions are the commonest. *Haemangioma* of the parotid in infants, usually girls: soft, bluish, enlarging with crying, growing in the first months then involuting by about five to seven years; half have skin haemangiomas; propranolol, surgery only if it fails. *Lymphangioma* does not involute and is excised or sclerosed. Among solid tumours in children, pleomorphic adenoma is the commonest benign one, but up to half are malignant, mostly *mucoepidermoid carcinoma*.

3 CHAPTER THREE Malignant tumours

By type.

3.1 Mucoepidermoid carcinoma

The commonest salivary malignancy in adults and children, usually in the parotid: a mixture of mucous, epidermoid and intermediate cells.

- **Low grade:** more mucous cells, cystic, slow, behaving almost like a benign tumour; five-year survival about 90%.
- **High grade:** more epidermoid cells, solid, aggressive, with nodal spread; survival about 30 to 50%.

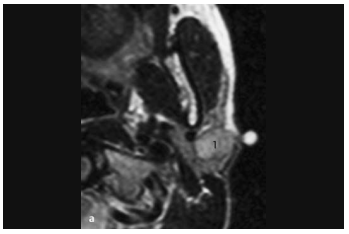
Low-grade tumours: parotidectomy (superficial or total according to site) preserving the nerve. High-grade: total parotidectomy, sacrificing only nerve that is invaded, with neck dissection and postoperative radiotherapy.

3.2 Adenoid cystic carcinoma (cylindroma)

About 10% of salivary cancers, the commonest in the submandibular gland, sublingual gland and minor glands (especially the palate). Slow but relentless: it infiltrates widely along tissue planes and especially *nerves* (perineural spread, causing pain, facial palsy and skip lesions to the skull base), recurs locally decades later, and metastasises to the *lungs* (often with long survival) more than to nodes. Histology: cribriform ("Swiss cheese", best prognosis), tubular, and solid (worst). Treatment: wide excision with clear nerve margins and postoperative radiotherapy for almost all; survival keeps falling beyond five and ten years, so follow-up is lifelong.

IMAGING IN THE APP 1 case

Adenoid cystic carcinoma on imaging.



Adenoid Cystic Carcinoma

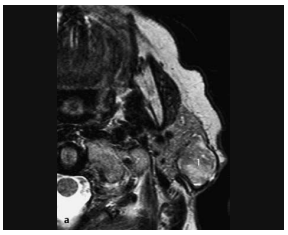
Tumours · 3 images

3.3 Acinic cell carcinoma

Usually in the parotid, sometimes bilateral, mostly in women: a small, firm, mobile lump resembling an adenoma; usually low grade. Parotidectomy; recurrence more often local than regional.

IMAGING IN THE APP 1 case

Acinic cell carcinoma on imaging.



Acinic Cell Carcinoma
Tumours · 3 images

3.4 Carcinoma ex pleomorphic adenoma

A carcinoma (often salivary duct carcinoma) arising in a long-standing or recurrent mixed tumour, up to 10% of salivary cancers: a sudden phase of rapid growth, pain or palsy. Radical surgery and radiotherapy; poor prognosis. (Rarer forms: true carcinosarcoma, and the metastasising pleomorphic adenoma.)

3.5 Salivary duct carcinoma

A high-grade cancer resembling breast ductal carcinoma, in older men, with early nodal and distant spread; often HER2 or androgen receptor positive, which allows targeted treatment of recurrence.

3.6 Others

Polymorphous adenocarcinoma (palate; good prognosis), epithelial–myoepithelial carcinoma, secretory carcinoma (ETV6–NTRK3 fusion, treatable with TRK inhibitors), adenocarcinoma NOS, and undifferentiated carcinoma (rapid, painful, fixed, with facial palsy and nodes).

3.7 Squamous cell carcinoma

Primary salivary squamous carcinoma is rare; most squamous cancer in the parotid is *metastatic* to intraparotid nodes from skin cancer of the scalp, ear, temple and face, which spreads to the parotid in about 5% of high-risk cases. Rapid growth, pain, ulceration, nodes. Radical parotidectomy with excision of skin, muscle,

mandible or temporal bone as needed, neck dissection, and postoperative radiotherapy.



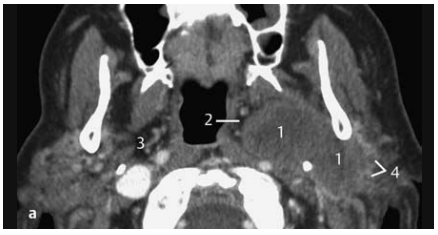
FIGURE 2 Squamous cell carcinoma of the right parotid: (A) a parotid swelling and (B) facial palsy.
Dhingra, fig. 46.1

3.8 Lymphoma and metastases

Primary MALT lymphoma of the parotid (especially in Sjögren syndrome) and secondary nodal lymphoma: biopsy and treat as lymphoma. Melanoma from the scalp and face also metastasises to the parotid.

IMAGING IN THE APP 1 case

Malignancies of the parotid gland.



Malignancies of the Parotid Gland
Tumours · 10 images

4 CHAPTER FOUR Treatment principles

Surgery first.

- **Primary tumour:** complete excision. In the parotid, superficial, total or radical parotidectomy according to site and extent; in the submandibular gland, excision of the gland with the contents of the triangle; minor gland cancers by wide local excision (often partial maxillectomy for the palate).
- **Facial nerve:** preserved whenever it can be separated from tumour, even with malignancy, provided it functions before surgery; a nerve that is paralysed or encased is resected and grafted immediately (great auricular or sural nerve), with mastoidectomy if needed to reach a clear proximal margin.
- **Neck:** therapeutic dissection (levels I to V) for clinical nodes; elective dissection for high-grade tumours, T3 and T4 tumours, extraglandular extension and facial nerve involvement.
- **Postoperative radiotherapy:** high-grade histology, close or positive margins, perineural invasion, extraglandular extension, nodes, and recurrent tumours; neutrons and protons have special roles.
- **Systemic treatment:** for metastatic disease; targeted agents for HER2, androgen receptor and NTRK-driven tumours.

T STAGING OF MAJOR SALIVARY GLAND CANCER (AJCC)

CATEGORY	DEFINITION
T1	2 cm or less, no extraparenchymal extension
T2	Over 2 cm up to 4 cm, no extraparenchymal extension
T3	Over 4 cm, or extraparenchymal extension
T4a	Skin, mandible, ear canal or facial nerve
T4b	Skull base, pterygoid plates, or encasing the carotid

5 CHAPTER FIVE Parotidectomy

The operation and its sequelae.

5.1 Technique

A modified Blair (lazy S, preauricular-cervical) incision or a facelift incision; a skin flap raised superficial to the parotid fascia; the great auricular nerve identified (its posterior branch preserved where possible); the gland mobilised off the ear cartilage and sternomastoid; the *facial nerve trunk* found using the tragal pointer, tympanomastoid suture and posterior belly of digastric, then followed forwards along its branches, lifting the superficial lobe off them. Nerve monitoring is widely used. Retrograde dissection from a peripheral branch helps in revision surgery.

OPERATIVE GUIDES IN THEATRE 2 operations

Superficial and deep lobe parotidectomy.

Superficial Parotidectomy
Superficial parotidectomy
Deep Lobe Parotidectomy
Deep lobe parotidectomy

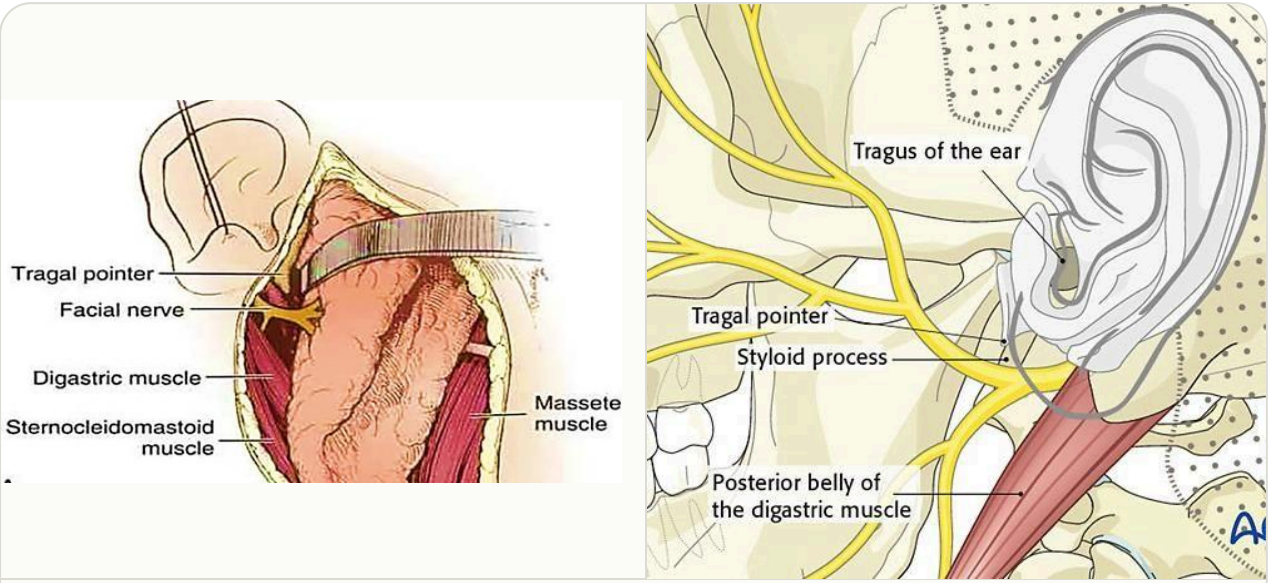


FIGURE 3 Landmarks for the facial nerve trunk: tragal pointer, styloid process and posterior belly of digastric.

Anatomy atlas · ENT Steps p. 185, ENT Steps p. 186

5.2 Complications

- **Facial nerve weakness:** temporary in a fifth to a third (especially the marginal mandibular branch), permanent in under 1 to 2% in benign surgery.
- **Numbness** of the ear lobe (great auricular nerve).

- **Frey syndrome (gustatory sweating):** sweating and flushing of the cheek skin while eating, months after surgery, because cut parasympathetic secretomotor fibres to the parotid regenerate into the sympathetic nerves of the sweat glands. Confirmed by the starch–iodine (Minor) test. Most need only reassurance; botulinum toxin injected into the skin is effective; an interposed barrier (SMAS flap, sternomastoid flap, acellular dermis, fat) reduces it; tympanic neurectomy is an old remedy.
- **Sialocele and salivary fistula** (usually settle with pressure, aspiration or botulinum toxin), haematoma, infection, a hollow contour, and tumour recurrence.

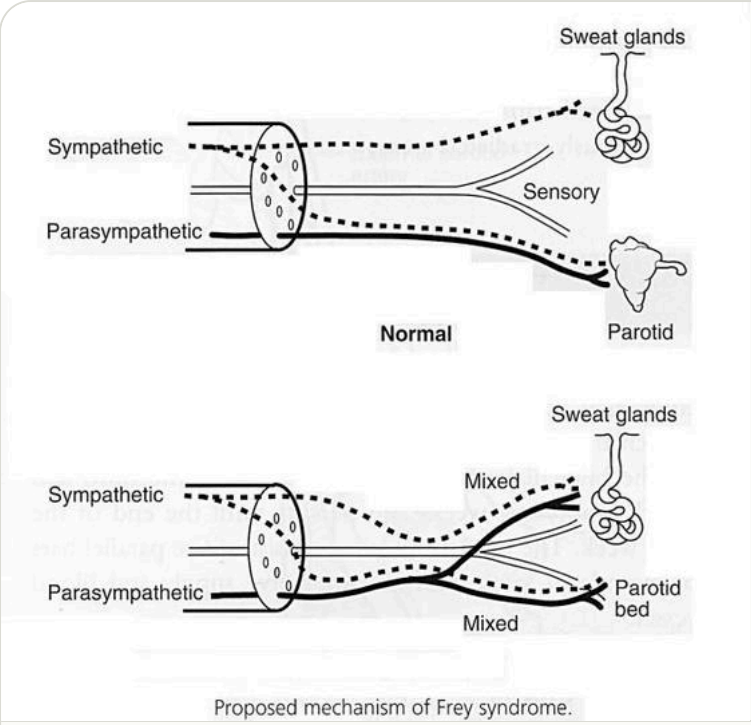


FIGURE 4 The mechanism of Frey syndrome.

Anatomy atlas · ENT Steps p. 188

NUMBERS FOR CONSENT (OXFORD TEXTBOOK)

The parotid holds 70 to 85% of salivary tumours, of which about 20% are malignant; the submandibular gland 8 to 15% (about 40% malignant); and minor and sublingual glands the rest, with about 80% malignant. Malignancies are graded low, intermediate or high, and high grade predicts local recurrence and distant spread. Fine-needle aspiration has a sensitivity and specificity approaching 90%, but up to 30% of results are indeterminate, which is where the **Milan system** for reporting risk, and core biopsy, help.

After parotidectomy, temporary facial weakness occurs in up to about a third (lower, about 22%, with continuous nerve monitoring) and permanent weakness in about 7%, which monitoring does not change. **Frey syndrome** (gustatory sweating and flushing from misdirected parasympathetic fibres) follows about 20% of superficial

parotidectomies; **first bite syndrome**, a sharp pain with the first mouthful of each meal from loss of sympathetic supply, follows deep lobe and parapharyngeal surgery. For the submandibular gland, the incision is placed 3 to 4 cm below the mandible, and the marginal mandibular nerve is protected by raising the fascia over the gland (the Hayes Martin manoeuvre) or by finding the nerve directly. A modified Blair incision is used whenever a neck dissection may be needed.

CHAPTER SIX

6 For the examination

What the examiner wants to hear.

6.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **The rule of glands:** larger gland, more likely benign.
- **Pleomorphic adenoma:** pathology, why not to enucleate, superficial parotidectomy, malignant change.
- **Warthin tumour:** who, where, bilaterality, pertechnetate.
- **Mucoepidermoid carcinoma** grades; **adenoid cystic carcinoma** and perineural spread.
- **Features of malignancy;** FNA versus open biopsy.
- **Parotidectomy:** facial nerve landmarks, complications, Frey syndrome.

6.2 Common pitfalls

COMMON PITFALLS

- Enucleating a pleomorphic adenoma.
- Doing an open biopsy of a parotid lump.
- Removing a parapharyngeal deep lobe tumour through the mouth.
- Sacrificing a working facial nerve that could be dissected free.

Written for this app from Oxford Textbook of Otolaryngology ch. 58, Oxford Specialist Handbook ch. 7, Oxford Handbook of ENT ch. 11, Dhingra ch. 46 and K. J. Lee ch. 35. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

SALIVARY GLANDS · TUMOURS

Pleomorphic Adenoma of the Parotid Gland

IMAGING FINDINGS

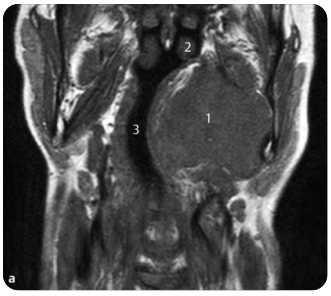
- Well-defined, lobulated mass with a capsule
- Very bright on T2
- Deep lobe: widens the stylomandibular gap, pushes parapharyngeal fat medially

TOP DIFFERENTIALS

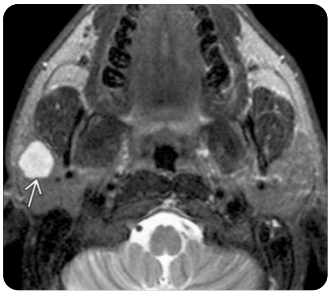
- Warthin tumour (tail, multiple, smokers)
- Low-grade mucoepidermoid carcinoma
- Intraparotid node

CHECKLIST

- Superficial or deep to the retromandibular vein
- Never enucleate: recurrence



MRI, T1-weighted, coronal. Note the extensive, isointense, smooth-bordered lesion (1) in the hypopharyngeal area: from the level o…



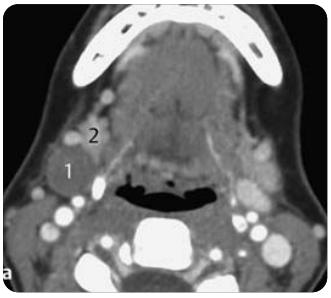
Axial T2 FS MR reveals a sharply circumscribed, high-signal BMT in the superficial lobe of the parotid gland. This tumor at the very least abuts, if…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 88.

SALIVARY GLANDS · TUMOURS

Pleomorphic Adenoma of the Submandibular Gland

- Calcifications are well depicted on CT, as are signs of infection and abscess formation. On CT, most pleomorphic adenomas are smooth-bordered, spherical, benign-looking tumors that usually display a higher density than the surrounding parotid parenchyma. Sometimes, these tumors demonstrate a heterogeneous appearance with sites of lower attenuation representing areas of necrosis, old hemorrhage, and/or cystic change (see
- For MRI characteristics, see Pleomorphic Adenoma of the Parotid Gland, page 316.



Patient with an asymptomatic slowly progressive swelling in the right submandibular region. Shaving his beard off revealed this pathology. CT…

SALIVARY GLANDS · TUMOURS

Malignancies of the Parotid Gland

IMAGING FINDINGS

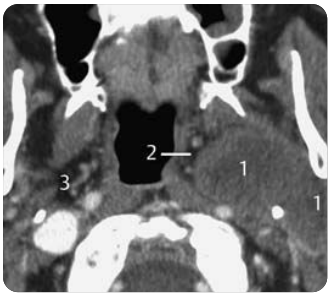
- Ill-defined or infiltrative margins
- Low or intermediate T2, low ADC
- Facial nerve: enhancement through the stylomastoid foramen

TOP DIFFERENTIALS

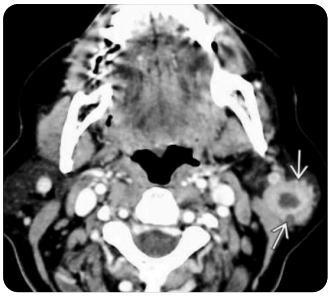
- Pleomorphic adenoma
- Warthin tumour
- Metastatic node from skin cancer; lymphoma

CHECKLIST

- Perineural spread to the mastoid segment and beyond
- Nodes in the parotid and level II



Slight swelling of the left parotid region and deep pain.
Dunnebier, Imaging for Otolaryngologists, Fig. 9.30 a



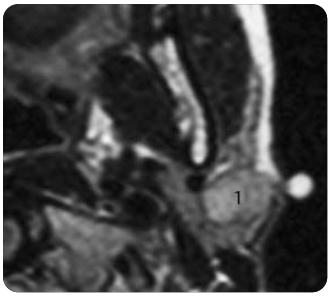
Axial CECT reveals a well-defined mass in the superficial lobe of the parotid gland. It has a thick rind of peripheral enhancement and is centrally cystic …

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 98.

SALIVARY GLANDS · TUMOURS

Adenoid Cystic Carcinoma

- Slow-growing but relentless, with perineural spread along VII (to the stylomastoid foramen and geniculate ganglion) and V3 (foramen ovale) as its hallmark; skip lesions are common.
- MRI with fat-saturated post-contrast T1 shows nerve thickening and enhancement and widening of the foramina; image the whole nerve to the brainstem.
- Late lung metastases are common; long follow-up (over 10 years) is needed.

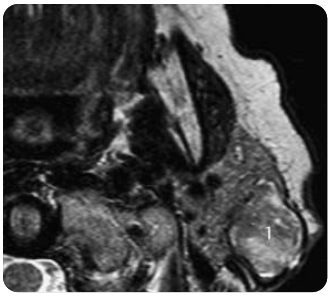


MRI, T2-weighted. Limited lesion within the boundaries of the parotid gland. The T2-weighted image (1) demonstrates a uniform high-to-…

SALIVARY GLANDS · TUMOURS

Acinic Cell Carcinoma

- A low-grade malignancy, mostly in the parotid, often well circumscribed and so easily mistaken for a benign tumour on imaging.
- May be cystic or multifocal; up to a few per cent are bilateral.
- Treat with parotidectomy; nodal disease is uncommon in low-grade tumours.

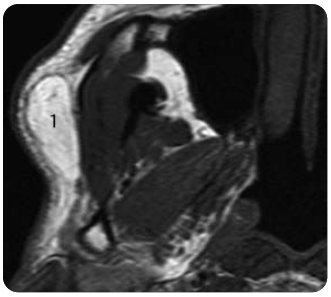


MRI, T2-weighted without gadolinium enhancement, coronal. Heterogeneous signal intensity (1) without evident cystic structures.…

SALIVARY GLANDS · TUMOURS

Lipoma Near the Parotid Gland

- Lipomas have fat density on CT (−50 to −100 HU) and follow fat signal on every MRI sequence, with suppression on fat-saturated images.
- Thin septa are acceptable; thick enhancing septa or nodules suggest liposarcoma.
- A soft, mobile, slowly growing mass; excision only for growth or cosmesis, with the facial nerve identified if it lies within the gland.



Asymptomatic weak swelling in the parotid region.

Dunnebier, Imaging for Otolaryngologists, Fig. 9.27 a